



Introduction to Statistics

IF2120 Probabilitas dan Statistika
Semester 1 – 2025/2026
Prodi Teknik Informatika - ITB



Chapter 1: Introduction to Statistics



Statistics & Probability

Source: UCI Department of Statistics:

- Statistics is the science concerned with developing and studying methods for collecting, analyzing, interpreting and presenting empirical data.
- Two fundamental ideas in the field of statistics are uncertainty and variation.
- Probability is a mathematical language used to discuss uncertain events and probability plays a key role in statistics. Any measurement or data collection effort is subject to a number of sources of variation.

Variables

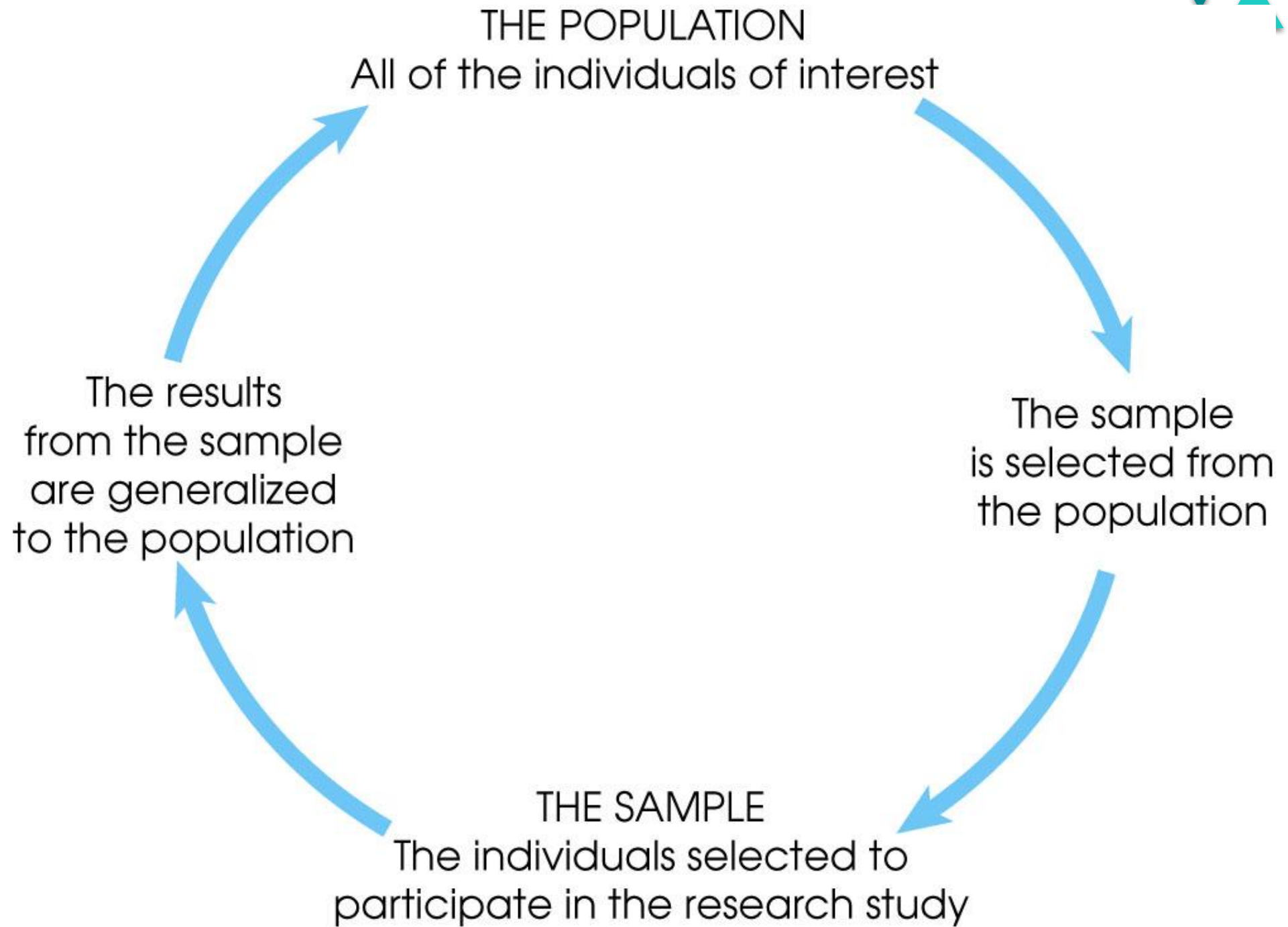
- A **variable** is a characteristic or condition that can change or take on different values.
- Most research begins with a general question about the relationship between two variables for a specific group of individuals.

Population

- The entire group of individuals is called the **population**.
- For example, a researcher may be interested in the relation between class size (variable 1) and academic performance (variable 2) for the population of third-grade children.

Sample

- Usually populations are so large that a researcher cannot examine the entire group. Therefore, a **sample** is selected to represent the population in a research study. The goal is to use the results obtained from the sample to help answer questions about the population.



Types of Variables

- Variables can be classified as discrete or continuous.
- **Discrete variables** (such as class size) consist of indivisible categories, and
- **Continuous variables** (such as time or weight) are infinitely divisible into whatever units a researcher may choose. For example, time can be measured to the nearest minute, second, half-second, etc.

Real Limits

- To define the units for a continuous variable, a researcher must use **real limits** which are boundaries located exactly half-way between adjacent categories.

Measuring Variables

- To establish relationships between variables, researchers must observe the variables and record their observations. This requires that the variables be **measured**.
- The process of measuring a variable requires a set of categories called a **scale of measurement** and a process that classifies each individual into one category.

4 Types of Measurement Scales

1. A **nominal scale** is an unordered set of categories identified only by name. Nominal measurements only permit you to determine whether two individuals are the same or different.
2. An **ordinal scale** is an ordered set of categories. Ordinal measurements tell you the direction of difference between two individuals.

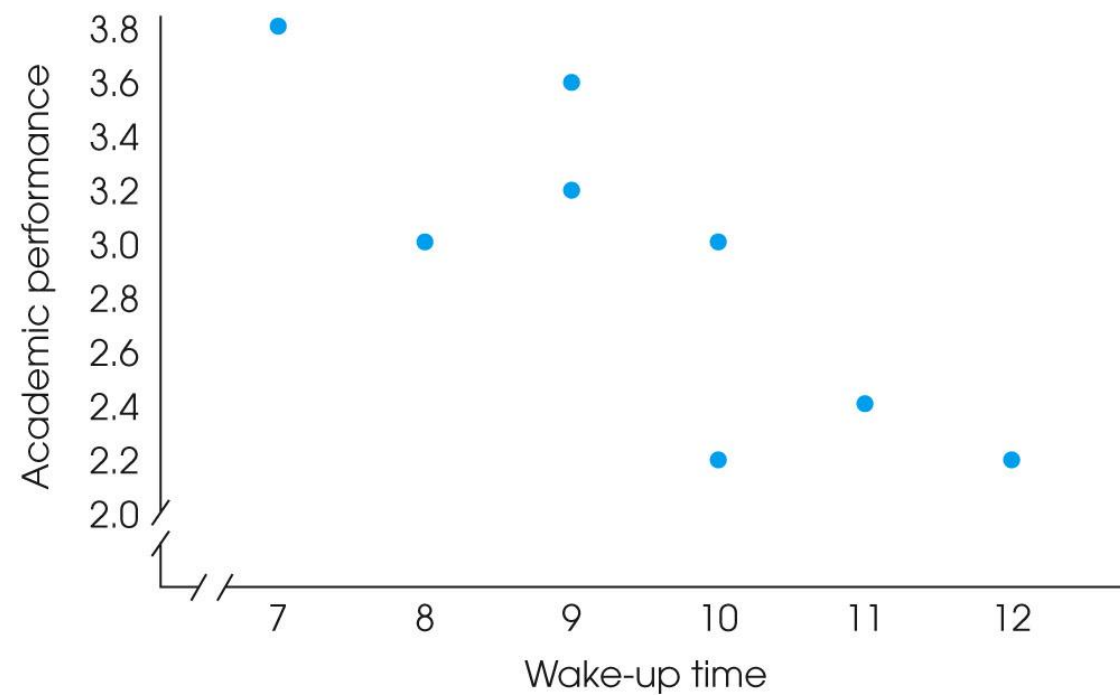
4 Types of Measurement Scales

3. An **interval scale** is an ordered series of equal-sized categories. Interval measurements identify the direction and magnitude of a difference. The zero point is located arbitrarily on an interval scale.
4. A **ratio scale** is an interval scale where a value of zero indicates none of the variable. Ratio measurements identify the direction and magnitude of differences and allow ratio comparisons of measurements.

Correlational Studies

- The goal of a **correlational** study is to determine whether there is a relationship between two variables and to describe the relationship.
- A **correlational** study simply observes the two variables as they exist naturally.

Child	Wake-up Time	Academic Performance
A	11	2.4
B	9	3.6
C	9	3.2
D	12	2.2
E	7	3.8
F	10	2.2
G	10	3.0
H	8	3.0



Experiments

- The goal of an **experiment** is to demonstrate a cause-and-effect relationship between two variables; that is, to show that changing the value of one variable causes changes to occur in a second variable.

Experiments (cont.)

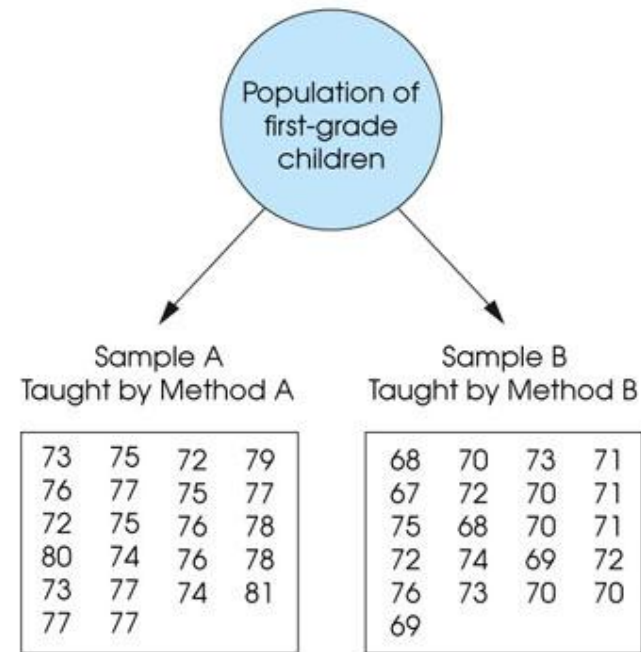
- In an **experiment**, one variable is manipulated to create treatment conditions. A second variable is observed and measured to obtain scores for a group of individuals in each of the treatment conditions. The measurements are then compared to see if there are differences between treatment conditions. All other variables are controlled to prevent them from influencing the results.
- In an experiment, the manipulated variable is called the **independent variable** and the observed variable is the **dependent variable**.

Step 1

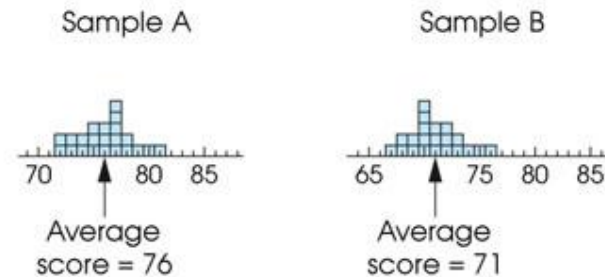
Experiment:
Compare two
teaching methods

Data

Test scores for the
students in each
sample

**Step 2**

Descriptive
statistics:
Organize and simplify

**Step 3**

Inferential
statistics:
Interpret results

The sample data show a 5-point difference between the two teaching methods. However, there are two ways to interpret the results:

1. There actually is no difference between the two teaching methods, and the sample difference is due to chance (sampling error).
2. There really is a difference between the two methods, and the sample data accurately reflect this difference.

The goal of inferential statistics is to help researchers decide between the two interpretations.

Descriptive Statistics

- **Descriptive statistics** are methods for organizing and summarizing data.
- For example, tables or graphs are used to organize data, and descriptive values such as the average score are used to summarize data.
- A descriptive value for a population is called a **parameter** and a descriptive value for a sample is called a **statistic**.

Inferential Statistics

- **Inferential statistics** are methods for using sample data to make general conclusions (inferences) about populations.
- Because a sample is typically only a part of the whole population, sample data provide only limited information about the population. As a result, sample statistics are generally imperfect representatives of the corresponding population parameters.

Other Types of Studies

- Other types of research studies, known as **non-experimental** or **quasi-experimental**, are similar to experiments because they also compare groups of scores.
- These studies do not use a manipulated variable to differentiate the groups. Instead, the variable that differentiates the groups is usually a pre-existing participant variable (such as male/female) or a time variable (such as before/after).

Other Types of Studies (cont.)

- Because these studies do not use the manipulation and control of true experiments, they cannot demonstrate cause and effect relationships. As a result, they are similar to correlational research because they simply demonstrate and describe relationships.

Variable #1: Subject gender
(the quasi-independent variable)
Not manipulated, but used
to create two groups of subjects

Variable #2: Verbal test scores
(the dependent variable)
Measured in each of the
two groups

Boys	Girls
17	12
19	10
16	14
12	15
17	13
18	12
15	11
16	13

Any
difference?

Variable #1: Time
(the quasi-independent variable)
Not manipulated, but used
to create two groups of scores

Variable #2: Depression scores
(the dependent variable)
Measured at each of the two
different times

Before Therapy	After Therapy
17	12
19	10
16	14
12	15
17	13
18	12
15	11
16	13

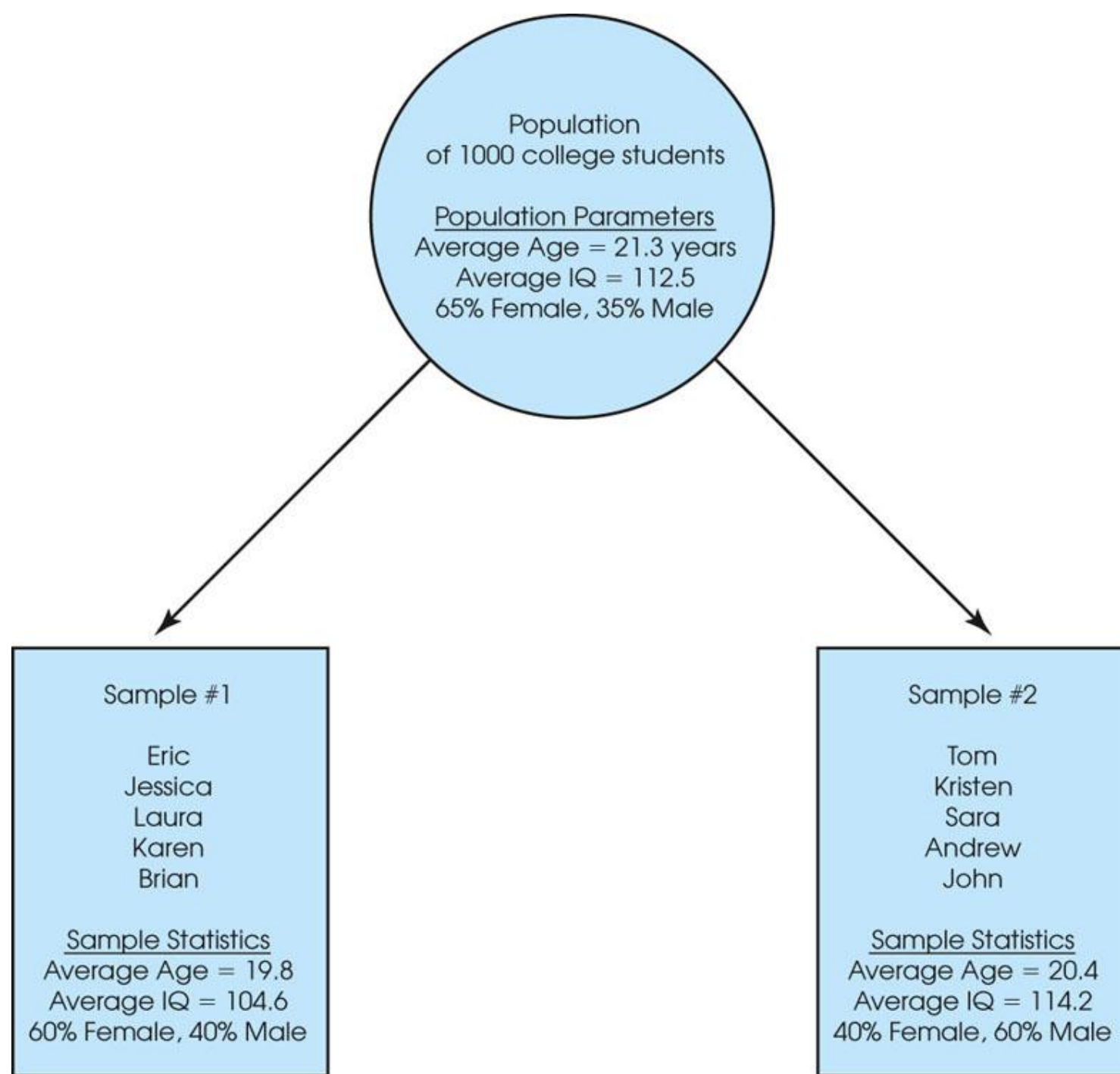
Any
difference?

Data

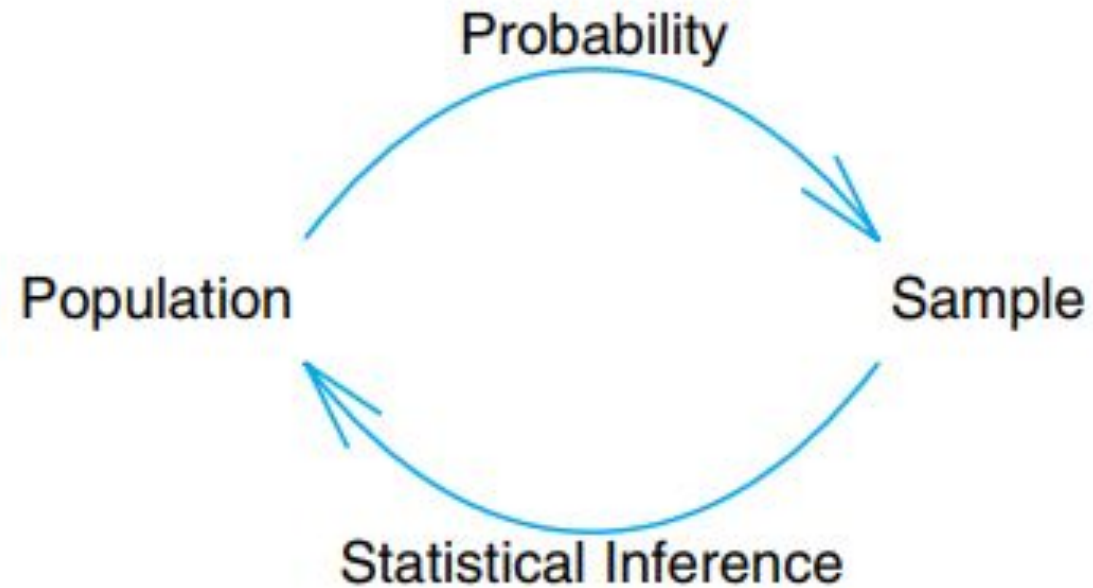
- The measurements obtained in a research study are called the **data**.
- The goal of statistics is to help researchers organize and interpret the data.

Sampling Error

- The discrepancy between a sample statistic and its population parameter is called **sampling error**.
- Defining and measuring sampling error is a large part of inferential statistics.



Probability and Statistics (work together)





**Any
questions?**



Thank you